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11.1 Care Guidelines

NOTE: **The Neonatal guidance is intended for the Neonatal Specialty Teams and is included here for reference and assistance should a generalist crew find themselves treating a neonate out of hospital. Initiation of these guidelines must be under the direction of online medical control for generalist crews.**

11.1.1 Pre-Departure Considerations

Based on type of neonatal patient and report received and/or specific medical direction, consider adding any of the following to the standard neonatal transport equipment and supplies:

- Curosurf (refrigerated)
- Alprostadil (refrigerated)
- Inhaled Nitric Oxide
- Additional IV pumps

11.1.2 Patient Assessment and Management Prior to and During Transport

Primary Assessment—perform a focused, rapid assessment at the time of patient rendezvous to identify immediate threats to life:

- Airway
- Breathing
- Circulation
- A secondary in-depth head to toe assessment should be performed when practicable.

Advanced Airway

If an advanced airway is in place, assess that it is secure and patent

- Place waveform capnography and assess waveform and EtCO2
- Nelcor pulse oximeter should be used if using Capnostream
- Suction as needed, except if surfactant was given within the last 6 hours
- Assess CXR forETT placement
- Assess current respiratory support or ventilator settings and effectiveness
- Assess current blood gas

Vital Signs

A set of vital signs should be obtained within 5 minutes of patient rendezvous.

- SpO2—consider Pre and Post ductal
- Heart rate
- Blood pressure—cuff or arterial
- Transducer should be placed on arterial lines
- Blood pressures that have been stable greater than 24 hours, with no history of hypotension, cardiac, or renal disease may be checked every 30 minutes
- Respiratory rate
- Temperature—place skin temperature probe for continuous monitoring
- N-PASS pain assessment

Repeat vital signs

- Every 15 minutes during transport, more frequent if dynamic
- For back transport of stable infants, every 30 minutes
- Minimum of 3 sets of vital signs on all transports

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11.1.3 Oxygen

Oxygen—Neonatal

Neonatal

- Undifferentiated—SpO2 90-98% based on patient condition or physician order
- Cardiac—SpO2 75-85% pre and post ductal
- Pulmonary Hypertension—SpO2 94-98% pre and post ductal
- Gestation less than 34 weeks— SpO2 88-95%

NOTE: **Administer and titrate oxygen to treat cyanosis or desaturation**

NOTE: **Wean oxygen gradually for high SpO2**

11.1.4 Venous Access and IV Fluid Management

Establish or assess venous access by peripheral vein or umbilical venous catheter unless infant has been stable on full feeds for greater than 24 hours

- Maintenance IV fluids D10W 80ml/kg/day
- UAC IV fluids at 0.5ml/hr-1ml/hr to maintain patency, heparinized if possible
- If available, assess x-ray for umbilical line placement or obtain report of placement

Glucose

- Obtain a bedside glucose based on infant's condition and history.
- Obtain glucose for infants less than 24 hours old with no glucose drawn within 30 minutes of rendezvous.
- No glucose check is necessary for asymptomatic infants that have been euglycemic for greater than 24 hours and are on established IV fluids or full feed.

11.1.5 Hypoglycemia

Dextrose

Neonatal

- 0.2 gm/kg (2 ml/kg D10W) IV over 2-4 min
- Repeat bolus PRN
- Follow with a D10W infusion at 80 mL/kg/24 hours

NOTE: **Neonatal hypoglycemia = glucose < 50 gm/dL**

- Recheck glucose every 15–30 minutes until hypoglycemia resolved
- Consider increasing maintenance fluids to 100 ml/kg/day or increasing dextrose concentration if hypoglycemia persists after boluses
- Consult medical director for specific orders if hypoglycemia persists after boluses

Hypotension

Mean blood pressure less than gestational age, delayed cap refill, and/or poor perfusion

- Normal Saline bolus 10 mL/kg IV over 10–60 minutes depending on severity of hypotension and gestational age
- Repeat PRN
- Consult medical director as soon as possible

11.1.6 Temperature

Maintain skin temperature 36.5-37.5 °C axillary/skin

- Consider using chemical warming mattress for neonates 35 weeks or less
- Initiate re-warming for temperatures less than 36.5 °C
- Consider using chemical mattress for neonates with initial temperature of 36.3 °C or less

11.1.7 Pain Management and Sedation

- Utilize NPASS for pain/agitation/sedation assessments in transport
- Use non-pharmacological techniques: pacifiers, sucrose, and swaddling to comfort infants
- Contact medical control for specific sedation/pain management orders when indicated
- Consider sedation for procedures or concern for advanced airway tolerance

Fentanyl

- 0.5 -1 mcg/kg IV over 1-2 min, IM, NAS
- Repeat every hour PRN

  NOTE: **May use for analgesia or post intubation sedation**

Morphine

- 0.05-0.1 mg/kg IV

Midazolam

- 0.1–0.2 mg/kg IV

Alternately for emergent procedural sedation if no IV is in place

- 0.3 mg/kg NAS
- Repeat every 2-4 hrs PRN

  NOTE: **For infants displaying no signs or symptoms of shock or hypotension.**

Lorazepam—Anxiolysis

- 0.1 mg/kg IV

11.1.8 Gastric Decompression

If condition requires, place an orogastric tube, empty the neonate's stomach and leave in place for gastric decompression during transport

- Bag-mask ventilation prior to transport arrival
- Recent emesis
- Suspected GI disorder
- RAM cannula

11.1.9 Intake and Output

Monitor accurate Intake and Output including IV fluid volumes, gastric, or other tube output and include it in handoff report

11.1.10 Medical History

- Obstetrical History—obtain report on maternal health, EDC, prenatal care, labor, delivery and any circumstances or complications relevant to infant's current condition.
- Neonatal History—obtain report on birth (location, method, appearance of amniotic fluid, complications), APGAR's, weight, resuscitation required at delivery
- History of Present Condition—obtain report on history of clinical course, medications, fluids or feedings, treatments/procedures completed prior to arrival of team, pertinent labs and diagnostic imaging.

11.1.11 Medication Reconciliation

- Obtain report of all the medications the patient is taking or has received prior to rendezvous, include the drug name, dosage, frequency, route, and time of administration.

Medical Direction

Prior to leaving the sending facility, contact medical control to provide an update on patient's assessment, and clarify orders in the following circumstances:

- Infant's condition has deteriorated since the initial report
- Procedures such as intubation, umbilical line placement, or active treatment of pneumothorax are necessary
- If patient's conditions requires, perform procedure first and notify Medical Director as soon as possible
- Medications other than dextrose, antibiotics, emergent volume bolus or emergent pre-procedural sedation are needed
- FiO2 greater than 60%, CPAP, or mechanical ventilation is required

NOTICE

Medical control contact is not required for back transport of stable infants, transports for procedures or special testing, or well stabilized infants on minimal support. The transport team may use their discretion in these circumstances.

11.1.12 Emergency Resuscitative Treatment

NRP guidelines should be used for neonatal resuscitation and stabilization in transport

- Provide bag-mask ventilation for neonatal apnea or as needed with respiratory decompensation
- Perform intubation for resuscitation or if infant's condition warrants.
- Intubation may be preferred to treat Congenital Diaphragmatic Hernia
- i-gel may be used as an alternative airway
- Initiate chest compressions if bradycardia less than 60 bpm persists despite effective ventilation
- Administer medications per NRP guidelines
- If pneumothorax is strongly suspected confirm by x-ray or transillumination and refer to guideline for treatment
- If assessment reveals clinical seizures administer care based on seizure protocol

Epinephrine—Neonatal Resuscitation

Neonatal

- 0.02 mg/kg IV
- IV ***volume dose*** 0.2 mL/kg
  - 1 kg = 0.2 mL
  - 2 kg = 0.4 mL
  - 3 kg = 0.6 mL
  - 4 kg = 0.8 mL
- Rapid push, followed with 3 mL normal saline flush

- 0.1 mg/kg via ETT
- ETT ***volume dose*** 1 mL/kg
  - 1 kg = 1 mL
  - 2 kg = 2 mL
  - 3 kg = 3 mL
  - 4 kg = 4 mL
- May be used while establishing IV access

NOTE: **0.1 mg/mL concentration, repeat every 3-5 min**

11.1.13 Supplemental Medications

NOTE: If unable to administer, include in handoff report at receiving facility.

- Administer Vitamin K IM and Erythromycin Ointment to both eyes if not given after birth

Vitamin K (phytonadione)

- Less than 1500 gm—0.5 mg IM
- 1500 gm or greater—1 mg IM

Erythromycin

- Apply 0.5–2 cm ribbon of ointment in each lower conjunctival sac
- One dose at birth if not already given by sending facility staff

11.1.14 Transport Paperwork

Parents

- Speak with parents
- Obtain Consent from parents
- If infant stability allows, let parents interact with infant
- Provide receiving hospitals map and phone number to parents

Sending Hospital Paper Work

Obtain the following:

Copy for receiving hospital personnel

- Sending facility chart
- Labs
- Diagnostics

Transport paperwork from sending hospital

- RFT
- High Risk Perinatal Program (HRPP) Enrollment Form
- Face sheet

11.1.15 Additional Transport Considerations

- Ensure infant has identification band before departing sending facility
- Deliver handoff report to receiving hospital personnel taking over care of infant
- Deliver paperwork and leave draft chart with receiving facility

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11.2 Neonatal Conditions

11.2.1 Very Low Birth Weight Infant

This guideline applies to infants less than 32 weeks gestation or 1500 grams who are transported in the first 24 hours of life

Pre-Transport Planning

- Add Poractant Alfa to transport supplies
- Preheat isolette to 38 ºC and have chemical warming mattress and infant wrap readily available

Thermoregulation

- Maintain goal temperature between 36.5-37.5 ºC
- Monitor infant's temperature with axillary or skin temp probe
- If infant already in plastic wrap, keep in plastic wrap for transport
- If infant < 6 hours old and not already in a plastic wrap, place infant in wrap
- If infant < 36.5 ºC, place on activated chemical thermal mattress with one layer of blanket between mattress and infant.
- Once infant in isolette, avoid opening door and access infant through portholes only as necessary

⚠ CAUTION **Watch for hyperthermia (>37.5 ºC) and remove chemical mattress or adjust isolette temperature if indicated.**

Positioning

- Place infant supine and maintain head midline
- If isolette is not level, ensure that infant's head is at elevated end
- Infant should be nested with arms and legs flexed and supported
- Minimize leg or buttock raising

Ventilation and Oxygenation

- Pulse oximetry probe should be placed pre-ductally (on infant's right hand or wrist)
- Maintain goal oxygen saturations of 88-95%
- Goal pH > 7.25 with goal CO2 on blood gas 40-55

Fluids and Electrolytes

Glucose

- Maintain euglycemia
- Use D10W bolus as needed to treat hypoglycemia (< 50 gm/dL)
- If blood glucose is >200 gm/dL, consult with Medical Director

Dextrose

Neonatal

- 0.2 gm/kg (2 ml/kg D10W) IV over 2-4 min
- Repeat bolus PRN
- Follow with a D10W infusion at 80 mL/kg/24 hours

NOTE: **Neonatal hypoglycemia = glucose < 50 gm/dL**

IV Fluids

- Maintain total fluids between 80-100 ml/kg (may be adjusted as clinical condition warrants)

Umbilical Catheter

Umbilical Artery Catheter

- Appropriate umbilical artery fluid includes: normal saline, ½ normal saline, or ½ normal sodium acetate and should be heparinized with heparin 0.5 units/mL
- Rate for umbilical artery catheter should be between 0.5–1 ml/hr. This may be adjusted as needed and per Medical Director
- Umbilical artery catheter should be transduced for blood pressure throughout transport

Umbilical Venous Catheter

- Appropriate umbilical venous catheter fluids include D10, D5 and TPN.
- Umbilical venous catheter fluid should be heparinized with 0.5u/mL when practical

Noise and Vibration

- Utilize gel mattress to lessen vibration
- Use infant hat with padding over infant's ears to lessen noise levels

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11.2.2 Neonatal Seizures

A seizure in the neonatal period is an emergency that reflects potentially significant insults to the immature brain. Diagnostic and therapeutic interventions should be established promptly to prevent hypoxia, hyperthermia, hypoglycemia, hypercarbia, respiratory failure and metabolic acidosis.

Assessment and Management Prior to and During Transport

Obtain detail prenatal, labor and delivery history and note potential conditions that can cause neonatal seizures

- Hypoglycemia
- Hypoxic Ischemic Encephalopathy
- Intraventricular Hemorrhage
- Metabolic imbalance
- Trauma
- Meningitis
- Withdrawal
- Genetic
- Fever

Seizure Activity

If seizure activity observed—note the following:

- Time of onset and duration
- Involvement of body parts
- Types of movement, e.g. subtle, tonic, focal clonic, multifocal clonic
- Cyanosis

Seizure Treatment

- Administer oxygen as needed to maintain acceptable SpO2
- Check blood glucose to rule out hypoglycemia
- Consult Medical Director for specific orders for medications
- Anticipate orders for Phenobarbital

Phenobarbital

- 20 mg/kg IV over 10–15 min
- Repeat every 30 min PRN if seizures persist
- Maximum total dose 40 mg/kg

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11.2.3 Neonatal Gastro-Intestinal Disorders

Neonatal GI disorders include intestinal atresia, stenosis, obstructions, perforation, malrotation, Hirschprung, necrotizing enterocolitis, imperforate anus

Assessment Prior to Transport

Obtain targeted prenatal history (polyhydramnios, family history)

- Observe for signs of hypovolemia and hypoglycemia due to vomiting, poor feeding or NPO status
- Assess amount and properties of vomitus or gastric output via nasal or orogastric tubes
- Assess for abdominal distention and measure girth, compare with previous baseline measurement
- Evaluate for associated/additional malformations
- Evaluate lab data including CBC, electrolytes
- View radiologic studies

Treatment Prior to and During Transport

- Place infant NPO
- If signs of hypovolemia, obtain IV/UVC access
- Consider fluid bolus and maintenance fluids
- Place replogle/OG tube for decompression if not yet done
- Place indwelling tube to open gravity
- Perform active intermittent decompression with large syringe if indicated during transport with considerations for effects of altitude
- Consult with Medical Director for any further orders
- Consider antibiotic administration

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11.2.4 Abdominal Wall Defect

Gastroschisis

Abdominal wall defect lateral to umbilical cord

- Assess color, perfusion and characteristics of exposed bowel
- Determine if white peel is present which indicates inflammation

Omphalocele

Abdominal wall defect covered by peritoneum and amnion

- Assess color, perfusion and characteristics of exposed bowel or other abdominal contents
- Assess if membrane is intact or ruptured
- Assess for associated cardiac anomalies, GI abnormalities or wide tongue associated with Beckwith-WiedemannSyndrome

Treatment

Treatment for gastroschisis and omphalocele:

- Cover exposed viscera in a sterile bag
- Support patient and defect well during movement to and from isolette
- Position patient on side and support defect, may need to place blanket beneath defect for support

  NOTE: **For gastroschisis—if there is a right side defect place baby on right side to prevent tension on mesenteric vessels**
- Place Replogle tube for bowel decompression
- Avoid NIV to minimize bowel distention
- Administer IV fluids by peripheral IV (100-120ml/kg for gastroschisis and, if membrane is ruptured, for an oomphalocele)
- Consider antibiotics (typically ampicillin and gentamicin) prior to transport per medical direction
- Monitor temperature, intake/output, and perfusion

Bladder Exstrophy

Defect in which the bladder develops outside the fetus and may involve the urinary tract, genital tract, and pelvic musculoskeletal structures

Treatment

- Position supine
- Assess placement of cord clamp to make sure it is not on bladder, remove and place cord tie if indicated
- Cover defect with sterile plastic wrap (irrigate with sterile normal saline if dry)
- Administer IV fluids and antibiotics per Medical Direction
- Assess for additional abnormalities (cardiac, gastrointestinal, renal, musculoskeletal and neurological)

11.2.5 Transports of Infants for Hospice Care

This guideline applies to infants who are being transported from sending facilities to another facility, hospice/respite care center or home for hospice care.

Pre-Transport Planning

- Obtain report from infant's nurse or attending physician
- Determine the infant's relative stability/instability
- Determine current support and support required for transport

Infant Hospice Transport—Physician Orders form

- The Hospice Transport—Physician Orders form may be sent to the sending facility for preplanning or be completed at sending bedside if indicated by pre-transport assessment of stability

Bedside at Sending Facility

- Consult with the referring physician regarding the DNR status of the infant and if any life sustaining measures should be taken during transport to destination
- Have referring physician complete and sign Infant Hospice Transport – Physician Orders if infant has potential for deterioration during transport
- Clarify any questions prior to departing sending bedside
- If a DNR order is written and in the chart, obtain a copy
- Discuss transport plans and infant's parent(s) or legal representative if present
- Confirm directly or through sending facility staff that receiving location is prepared to receive the infant

Post-Transport Documentation

- Upload copy of DNR order to EHR
- Upload copy of Infant Hospice Transport – Physician Orders to EMR (if obtained)

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11.3 Neonatal Procedures

11.3.1 Neonatal Hypothermic Therapy

Indications

NOTICE

This procedure should be reserved for infants that have been identified as candidates for hypothermic therapy by the transport medical director, sending physician or receiving physician.

Possible candidates include infants 36 weeks gestation or greater with clinical evidence of moderate to severe hypoxic-ischemic encephalopathy and one or more of the following:

- Low APGAR score (< 5 at 10 minutes)
- Continued need for resuscitation (ET or mask ventilation) at 10 min of age
- Severe acidosis (pH <7.00 from cord gas or blood gas in first hour of life)
- Base excess (> 12 mmol/L from cord gas or blood gas in first hour of life)
- Moderate to severe encephalopathy (one or more of the following: lethargy, stupor or coma, hypotonia, abnormal reflexes, absent or weak suck, clinical seizures, abnormal EEG background and/or seizures)

Contraindications

- <36 weeks gestation
- Intrauterine growth restriction (IUGR) <1800 grams
- Severe congenital anomalies/syndromes/known metabolic disorders
- Evidence of head trauma

Procedure

Temperature Control

NOTICE

Actively cooling neonatal patients using cold packs or other cooling devices is prohibited.

CAUTION

Hyperthermia should be avoided

1. Continuously monitor temperature using an esophageal probe
2. Turn the isolette heating unit off or set it to the lowest setting possible
3. Maintain esophageal temperature 33.5 °C–34 °C
4. Esophageal temperatures below 33.5 °C should be avoided
5. If infant's temperature is below desired temperature, the isolette should be turned on at lowest setting needed to bring the infant's temperature back into the goal range of 33.5 °C–34 °C.

Special Considerations

IV Fluids

- Maintain IV fluids at 80 ml/kg/day
- If the sending facility has already started IV fluids at a rate greater than 80ml/kg/day, decrease rate to 80 ml/kg/day
- If the sending facility has already started fluids and the rate is 60–80 cc/kg/day, the rate may be left at this level for the duration of the transport

Respiratory Support

- The goal should be to normo-ventilate infants with goal pCO2 40–50. Values outside this range can have an effect on cerebral perfusion.
- Target arterial saturation should be 88%-98%
- Blood gas should be available for team before transport. However, the lack of a blood gas does not preclude the request for cooling evaluation.
- Blood gases should be obtained by the transport team per routine.
- Ventilation should be adjusted to maintain goal ventilation targets.

Medication Support

- Medications should be given as ordered by sending practitioner or transport medical director (as appropriate per medication guidelines)
- Shivering of the infant should be avoided, morphine sulfate can be used if the infant is shivering
- Avoid use of benzodiazepines
- Avoid sodium bicarbonate boluses unless specifically ordered by the receiving physician or transport medical director
- It is permissible to give ampicillin
- It is permissible to give first dose of gentamicin.

Gentamicin—Respiratory Infection, Sepsis, Open CNS or Abdominal Wall Defect

- 29 weeks or fewer—5.0 mg/kg
- 30-34 weeks—4.5 mg/kg
- 35 weeks or greater—4 mg/kg
- IV over 30 minutes via syringe pump
- IM acceptable if unable to establish IV access (undiluted)

Ampicillin

- 50 mg/kg IV over 3-5 minutes (preferred route), IM

  NOTE: **If meningitis is suspected, consult with medical control about higher dosing**

  NOTE: **May give IM if unable to establish IV access**

  NOTE: **100 mg/kg dose reserved for infants with positive signs of meningitis**

Neurological Support

- Observe infant for signs of seizure activity
- If infant develops clinical seizures, contact transport medical director for consideration of anticonvulsant therapy or other adjunct support measures

Special Considerations

Further medical direction is required if:

- Orders for treatment of seizures or shivering are needed
- An infant meets inclusion indicators and possibility of hypothermic therapy has not been addressed by transport medical director, referring physician or receiving physician

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11.3.2 TXP-2D Neonatal High Frequency Percussive Ventilator (HFPV)

- The TXP-2D is a pneumatically powered HFPV, it combines HFV and PCV as known as HFOV.
- The circuit used with the TXP-2D is the Phasitron Breathing Circuit, which delivers percussive pulses of air (at rates of 200-700 times per minute) while also delivering a continued wedge pressure to maintain constant alveolar inflation, prevent inflation-deflation cycle and improve oxygenation
- More efficient in clearing secretions and optimizing lung recruitment by maintaining a minimal mean intrathoracic expiratory pressure while minimizing potential barotrauma
- The TXP-2D has an active inspiratory phase and a passive exhalation phase
- Minimizes both regional lung over-distention (via barotrauma and volutrauma) and ventilatory-induced lung injury

Indications

- Transport of neonates already on hospital-initiated HFOV
- Failure of conventional mechanical ventilation

Contraindications

Absolute

- Untreated tension pneumothorax

Relative

- Pulmonary air leak
- Recent pneumonectomy
- Pulmonary hemorrhage
- Active emesis

Warnings

- TXP-2D with DM display (weight: ~2.76lb)
- Phasitron Breathing Circuit
- Medical gasses, oxygen, and air

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Pre-Use Check

Set up

1. Connect the Phasitron clear tubing to the “Fast Rate” port on the isolette
2. Connect the Phasitron red tubing to the “Gauge” port on the isolette
3. Ensure the black toggle is pointing to “High Frequency” mode on the isolette
4. Ensure both O_2 and air tanks are on
5. Turn on O_2 and air toggles on isolette with the Crossvent

Check Procedure

1. Rotate the “Frequency” knob to full clockwise position (turn all the way to the right)
2. Rotate the “Amplitude” knob to full clockwise position (turn all the way to the right)
3. Connect the Phasitron to the TXP-2D red and silver bulkhead fittings
4. Close the Phasitron patient port (with or without the HME) ensuring a tight seal
5. Rotate the “Amplitude” knob to full counterclockwise position (turn all the way to the left)
6. On the DM of the TXP-2D, observe the frequency on the bottom right side of the display is less than 250 pulses per minute and the MAP (mean airway pressure) on the center of the display is greater than 15cmH_2O
7. Open the Phasitron patient port and the disconnection alarm MUST sound
8. Re-close the Phasitron patient port and alarm should stop to continue pre-use check
9. Rotate the “Frequency” knob to full counterclockwise position (turn all the way to the left)
10. On the DM of the TXP-2D, observe the frequency on the bottom right side of the display is greater than 600 pulses per minute and the MAP on the center of the display is greater than 15cmH_2O
11. Rotate the “Frequency” knob to the 1200 position
12. Slowly rotate the “Amplitude” knob clockwise (turn to the right) and on the DM observe a decrease in the amplitude (on the upper right side of the display) until “off” at the full clockwise position
13. Pre-use check is complete

Initial Settings

NOTE: **Pre-Use Check already completed**

1. Place neo test lung on patient airway connector on Phasitron Breathing Circuit
2. Set Amplitude for adequate “chest wiggle” on a neonate, visual movement from shoulders to umbilicus: 15 – 25cmH_2O
3. Set MAP: 10–14 cmH_2O (match MAP on current HFV settings or go 1 – 2cmH2O above MAP on CMV) unless specific rate is applicable for known diagnosis
4. Set Frequency 500 pulses per minute (recommended range: 400–600 pulses per minute)
5. Set FiO_2

Adjust Settings

- Oxygenation (PaO2) – controlled by manipulating FiO2 and/or increasing PEEP and MAP; increased PaO2 can be achieved by increasing either the FiO2 or other parameters that directly influence MAP, such as increased PEEP, frequency and/or amplitude
- Ventilation (PaCO2) – controlled by manipulating delivered tidal volumes by changes in amplitude (directly related to tidal volume) and/or oscillatory frequency (i.e. changes in oscillatory frequency affect inspiratory time thereby influencing tidal volume

To Increase Oxygenation (↑ PaO2):

- Increase PEEP by 1–2 cmH2O
- Increase frequency by 60 pulses per minute
- Increase Amplitude 60 pulses per minute
- Increase FiO2

To increase ventilation (↓ pCO2):

- Increase amplitude by 2cmH2O (amplitude is directly related to tidal volume)
- Decreasing frequency by 60 pulses per minute (decreasing the oscillatory frequency leads to increased inspiratory time, thereby increasing tidal volume)

To Decrease Ventilation (↑pCO2):

- Decrease amplitude by 2cmH2O
- Increase frequency by 60 pulses per minute
- Decrease PEEP by 1 – 2cmH2O (may potentially alleviate overdistention of lung volumes, but patient's SpO2 must be monitored closely as this change may also negatively influence oxygenation)

Inhaled Nitric Oxide (iNO)

- A colorless gas that rapidly diffuses across the alveolar-capillary membrane and into adjacent smooth muscle of pulmonary vessels resulting in a selective pulmonary vasodilator
- Can decrease pulmonary vascular resistance and reduce ventilation-perfusion mismatch, both of which can contribute to improved PaO_2 and hemodynamic stability

Indications

- Need to improve oxygenation and reduce hypoxic respiratory failure associated with persistent pulmonary hypertension of the newborn and other conditions that may contribute to pulmonary hypertension such as respiratory distress syndrome, meconium aspiration syndrome, pneumonia, sepsis, cardiac anomalies, etc.

Precautions

- Rebound pulmonary hypertension – the abrupt discontinuation of iNO may lead to increased pulmonary artery vascular resistance and thus worsening oxygenation failure; if this occurs, reinstate iNO therapy immediately
- Hypoxemia from methemoglobinemia – nitric oxide (NO) combines with hemoglobin to form methemoglobin, which has decreased oxygen-carrying capacity due to fewer hemes to bind O_2; methemoglobin levels increase with iNO therapy
- Airway injury from nitric dioxide (NO_2) – NO and O_2, when mixed, forms NO_2 that may cause airway inflammation and damage lung tissue
- Worsening of severe left ventricular dysfunction – may cause pulmonary edema, increased pulmonary experience pulmonary edema, increased pulmonary capillary wedge pressure, systemic hypotension and bradycardia